Responsible AI Bias Detection Project

1. Objective

The objective of this project is to identify and analyze bias in AI-based decision-making systems. Responsible AI aims to ensure fairness, transparency, and accountability in automated decisions.

2. Dataset Description

A dataset containing 100 candidate records was created for this project. The dataset includes the following attributes:

- Gender
- Score
- Selected

The dataset consists of 50 male candidates and 50 female candidates.

	Gender	Score	Selected
0	Male	70	1
1	Male	71	1
2	Male	72	1
3	Male	73	1
4	Male	74	1

```
Gender
Male      50
Female    50
Name: count, dtype: int64
```

3. Tools Used

- Google Colab
- Python
- Pandas
- Matplotlib

4. Libraries Used

- Pandas
- Matplotlib

5. Data Preparation

The dataset was generated to simulate a recruitment scenario. Each record contains information about a candidate's gender, score, and selection status.

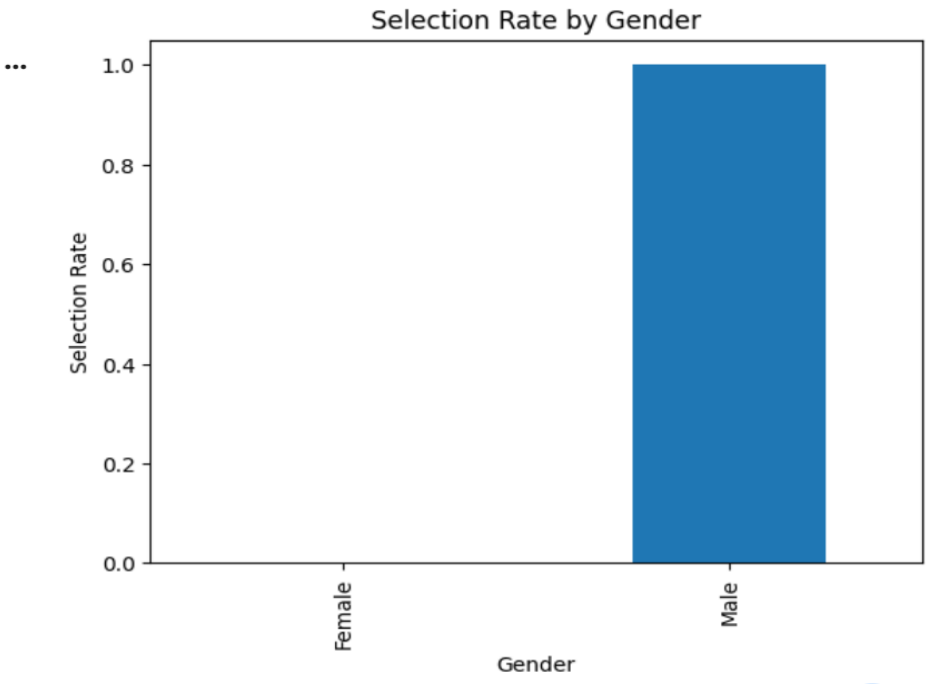
6. Bias Analysis

Bias occurs when decisions unfairly favor one group over another. In this dataset, candidates with similar scores receive different selection outcomes based on gender. The analysis was performed by calculating the average selection rate for each gender group.

```
Gender
Female    0.0
Male      1.0
Name: Selected, dtype: float64
```

7. Data Visualization

A bar chart was created to visualize the selection rate for male and female candidates. The chart clearly highlights the difference in selection outcomes.



8. Results

The analysis showed that male candidates had a higher selection rate compared to female candidates. This indicates the presence of gender bias in the decision-making process.

9. Responsible AI Considerations

To reduce bias in AI systems:

- Use balanced datasets.

- Monitor model predictions regularly.
- Apply fairness evaluation metrics.
- Ensure transparency in decision-making.
- Perform regular audits of AI systems.

10. Conclusion

This project demonstrated how bias can be identified using data analysis techniques. Responsible AI practices are essential to ensure fairness and ethical decision-making. Detecting and mitigating bias helps create trustworthy AI systems that treat all individuals fairly.